

Sachin Bala

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Hiring Team

Google, Operations Data Science

Austin, TX

Dear Hiring Manager,

I am writing to apply for the Product Data Scientist, Operations role at Google. With a background in data-driven process engineering and hands-on experience building analytics pipelines, statistical models, and real-time dashboards in high-volume manufacturing environments, I am excited to bring that same analytical rigor to Google's cloud infrastructure operations supporting Google Compute Engine and Vertex AI products.

At Tesla, I work at the intersection of operations and data science every day. I designed and deployed data collection and visualization pipelines using MySQL, Python, R, JMP, and Grafana to generate predictive insights into tool and process performance, directly enabling operational and financial decisions at the factory level. I applied DOE, SPC, and statistical capability analysis to large operational datasets, formulating quantifiable hypotheses, evaluating custom metrics, and delivering findings to engineering and business stakeholders. I also built a real-time Grafana monitoring pipeline for production tools, giving the team proactive visibility into equipment health before failures escalated -- the same kind of insight generation Google's operations team pursues for cloud reliability.

Earlier, I built a predictive failure-detection model using ML algorithms that improved manufacturing accuracy by 25%, and a Computer Vision defect detection system validated with Gauge R&R and F1-score analysis. Outside of work, I build AI software: my open-source project DataIQ (github.com/Sachin1bala1) is a Tauri + React desktop IDE where an AI agent autonomously queries databases, builds pipelines, and surfaces insights -- directly relevant to your team's goal of bringing AI into data science workflows.

Google's mission to make data work smarter for infrastructure operations aligns closely with the work I have been doing -- translating messy operational data into clear metrics, dashboards, and corrective actions. I am based in Austin, TX and eager to contribute to the GCE and Vertex product operations team.

Thank you for your time and consideration. I look forward to the opportunity.

Sincerely,

Sachin Bala